

2 Gibberellins are a group of plant hormones that are involved in the elongation of plant stems.

A student investigated the effect of two different concentrations of a gibberellin, known as GA₃, on stem elongation of 10-day old pea seedlings.

The student was given a stock solution of GA₃ with a concentration of $2.85 \times 10^{-4} \text{ mol dm}^{-3}$.

- (a) Describe a method the student could use to make a solution of GA₃ with a concentration of $1.90 \times 10^{-6} \text{ mol dm}^{-3}$ **and** state the dilution factor used.

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..... [2]

- (b) In the investigation, the student:

- used the two concentrations of GA₃: $2.85 \times 10^{-4} \text{ mol dm}^{-3}$ (high concentration GA₃) and $1.90 \times 10^{-6} \text{ mol dm}^{-3}$ (low concentration GA₃)
- applied the high concentration GA₃ to one batch of 10-day-old pea seedlings
- applied the low concentration GA₃ to another batch of 10-day-old pea seedlings
- applied distilled water to a third batch of 10-day-old pea seedlings, as a control
- standardised all other variables
- measured the length of the stem of each seedling every two days until the seedlings were 20 days old
- calculated the rate of stem elongation in cm day^{-1} .

State **one** way in which the student could standardise the measuring of stem length.

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(c) The results of the investigation are shown in Table 2.1.

Table 2.1

age of seedling / days	mean stem length with high concentration GA ₃ added / cm	mean stem length with low concentration GA ₃ added / cm	mean stem length with distilled water added / cm
10	2	2	2
12	4	2	3
14	7	5	4
16	6	12	7
18	20	15	8
20	25	21	8
rate of stem elongation / cm day ⁻¹	2.3		0.6

- (i) Complete Table 2.1 by calculating the rate of stem elongation in cm day⁻¹ for the seedlings with low concentration GA₃ added.

[1]

- (ii) A scientist stated that they did not have enough confidence in the results in Table 2.1 to make any conclusions.

State **one** reason to support the statement made by the scientist.

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 [1]

- (iii) Describe how the student could modify the investigation to increase confidence in the results.

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 [1]

- (d) The student found a study on the internet that considered the effect of different wavelengths of light on the concentration of two types of gibberellins, GA_1 and GA_8 , in pea plants.

In the study, 60 pea seeds were germinated and kept in the dark (no light) for seven days.

The pea seedlings were divided into four batches of 15:

- batch 1 remained in the dark
- batch 2 was exposed to blue light (470 nm)
- batch 3 was exposed to red light (680 nm)
- batch 4 was exposed to far-red light (750 nm).

After 4 hours the concentrations of GA_1 and GA_8 in the seedlings were measured.

Table 2.2 shows the results.

Table 2.2

batch	wavelength of light/nm	colour of light	mean GA ₁ concentration ± standard deviation /ng g ⁻¹ fresh mass	mean GA ₈ concentration ± standard deviation /ng g ⁻¹ fresh mass
1	–	dark	2.87 ± 0.72	3.07 ± 0.10
2	470	blue light	0.21 ± 0.05	3.90 ± 0.13
3	680	red light	0.31 ± 0.10	5.03 ± 0.21
4	750	far-red light	1.38 ± 0.26	4.00 ± 0.72

The student calculated the standard error (SE) and 95% confidence intervals (95% CI) for the data shown in Table 2.2. The formulae that the student used were:

$$SE = \frac{s}{\sqrt{n}}$$

$$95\% \text{ CI} = \bar{x} \pm (2 \times SE)$$

key to symbols

s = standard deviation

n = sample size (number of observations)

$\bar{x}$ = mean

Table 2.3 shows the calculated values for SE and 95% CI for the data from Table 2.2.

Table 2.3

batch	SE for GA ₁ /ng g ⁻¹ fresh mass	mean ± 95% CI for GA ₁ /ng g ⁻¹ fresh mass	SE for GA ₈ /ng g ⁻¹ fresh mass	mean ± 95% CI for GA ₈ /ng g ⁻¹ fresh mass
1	0.186	2.87 ± 0.37	0.026	3.07 ± 0.05
2	0.013	0.21 ± 0.03	0.034	3.90 ± 0.07
3	0.026	0.31 ± 0.05	0.054	5.03 ± 0.11
4			0.186	4.00 ± 0.37

Complete Table 2.3 to show the calculated values for SE and mean ± 95% CI for the data from far-red light for GA₁ (batch 4). [2]

(e) Fig. 2.1 shows the bar chart of the results from the study.

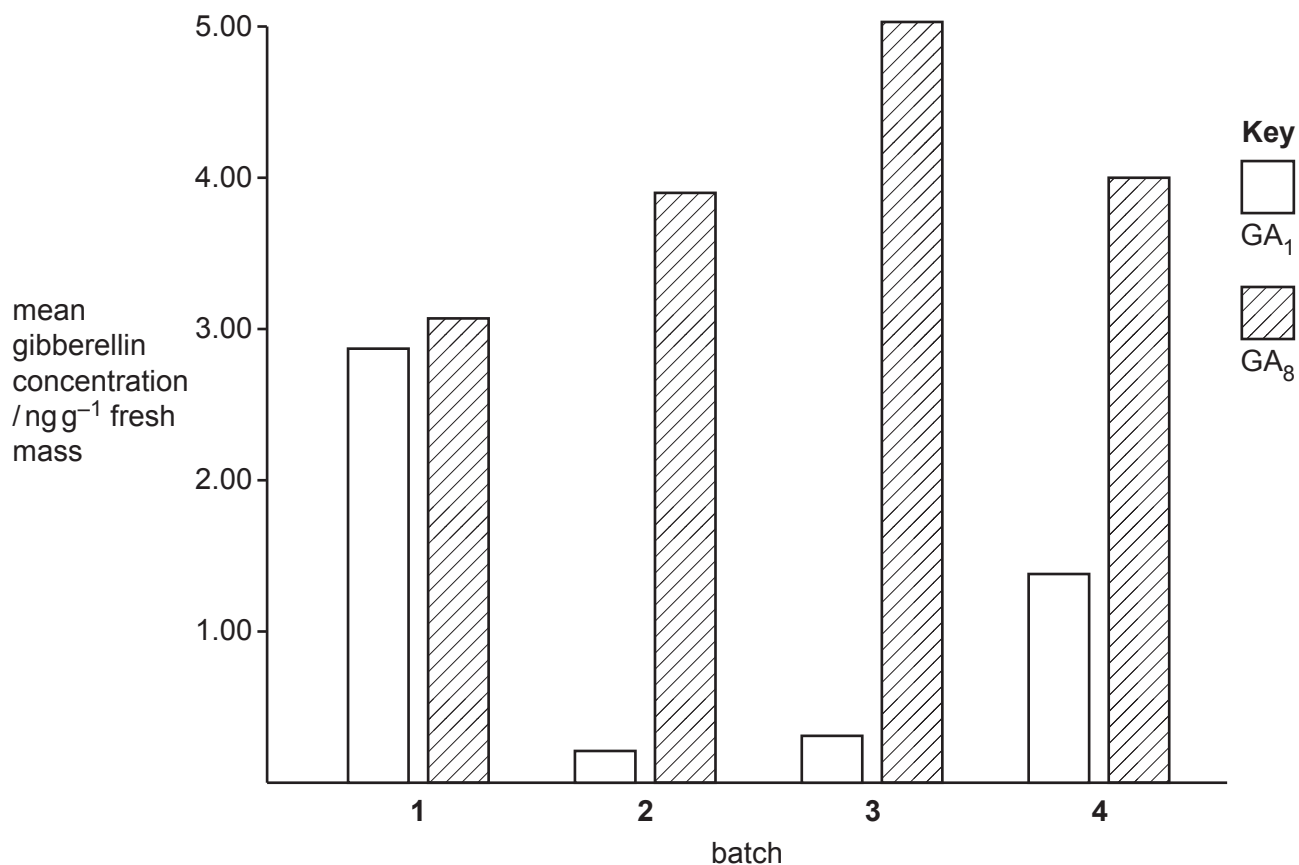


Fig. 2.1

Use Fig. 2.1, Table 2.2 **and** Table 2.3 to state **three** conclusions about the effect of different wavelengths of light on the concentrations of GA₁ and GA₈ in plants.

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[Total: 11]